

LABSHEET-9

Design and Development of a Full-Stack “Student Record Management” Web Application

1. server.js

```
const dns=require("dns");dns.setServers(["8.8.8.8","1.1.1.1"]);

const
express=require("express"),mongoose=require("mongoose"),cors=require("cors"),path=require("
path"),Student=require("./models/Student");

const
app=express();app.use(cors());app.use(express.json());app.use(express.static(path.join(__dirname
,"public")));

mongoose.connect("YOUR_MONGODB_ATLAS_CONNECTION_STRING").then(()=>console
.log("MongoDB Connected")).catch(e=>console.log(e.message));

app.post("/students",async(req,res)=>{try{res.status(201).json(await
Student.create({...req.body,marks:Number(req.body.marks)}))}catch(e){res.status(400).json({err
or:e.message})}}});

app.get("/students",async(req,res)=>res.json(await Student.find()));

app.get("/students/:id",async(req,res)=>res.json(await Student.findById(req.params.id)));

app.put("/students/:id",async(req,res)=>res.json(await
Student.findByIdAndUpdate(req.params.id,req.body,{new:true,runValidators:true})));

app.delete("/students/:id",async(req,res)=>{await
Student.findByIdAndDelete(req.params.id);res.json({message:"Deleted"})});

app.listen(5000,()=>console.log("Server running on 5000"));
```

2. models/Student.js

```
const mongoose=require("mongoose");

const schema=new mongoose.Schema({

  name:{type:String,required:true},rollNo:{type:String,required:true},

  course:{type:String,required:true},marks:{type:Number,min:0,max:100,required:true}});

module.exports=mongoose.model("Student",schema);
```

3. public/index.html

```
<!DOCTYPE html><html><head><title>Student CRUD</title><link rel="stylesheet"
href="style.css"></head>

<body><h1>Student Records</h1>

<form id="form"><input id="name" placeholder="Name" required><input id="rollNo"
placeholder="Roll No" required><input id="course" placeholder="Course" required><input
id="marks" type="number" min="0" max="100" placeholder="Marks"
required><button>Add</button></form>

<table
border="1"><thead><tr><th>Name</th><th>Roll</th><th>Course</th><th>Marks</th><th>A
ction</th></tr></thead><tbody id="table"></tbody></table>

<script src="script.js"></script></body></html>
```

4. public/script.js

```
const form=document.getElementById("form"),table=document.getElementById("table");

async function load(){let r=await fetch("/students"),d=await
r.json();table.innerHTML=d.map(s=>`<tr><td>${s.name}</td><td>${s.rollNo}</td><td>${s.co
urse}</td><td>${s.marks}</td><td><button onclick="edit('${s._id}')">Edit</button><button
onclick="del('${s._id}')">Delete</button></td></tr>`).join("")}

form.onsubmit=async e=>{e.preventDefault();let
d={name:name.value,rollNo:rollNo.value,course:course.value,marks:+marks.value;if(d.marks<
0||d.marks>100)return alert("Marks 0-100");await
fetch("/students",{method:"POST",headers:{"Content-
Type":"application/json"},body:JSON.stringify(d)});form.reset();load()};

async function del(id){await fetch("/students/"+id,{method:"DELETE"});load()}

async function edit(id){let s=await(await
fetch("/students/"+id)).json(),d={name:prompt("Name",s.name),rollNo:prompt("Roll",s.rollNo),
course:prompt("Course",s.course),marks:+prompt("Marks",s.marks)};await
fetch("/students/"+id,{method:"PUT",headers:{"Content-
Type":"application/json"},body:JSON.stringify(d)});load()}load();
```

5. public/style.css

```
body{font-family:Arial;text-
align:center;margin:30px}input,button{padding:7px;margin:4px}table{margin:20px auto;border-
collapse:collapse;width:80%}td,th{padding:8px}
```

OUTPUT

Student Record Management System

Student Records

Name	Roll No.	Course	Marks	Actions
Anshika	08	B. Tech	94	Edit Delete
Ansh	05	B. Tech	96	Edit Delete

localhost:5000 says
Enter student name:

Student Record Management System

Student Records

Name	Roll No.	Course	Marks	Actions
Anshika	08	B. Tech	94	Edit Delete
Ansh	05	B. Tech	96	Edit Delete